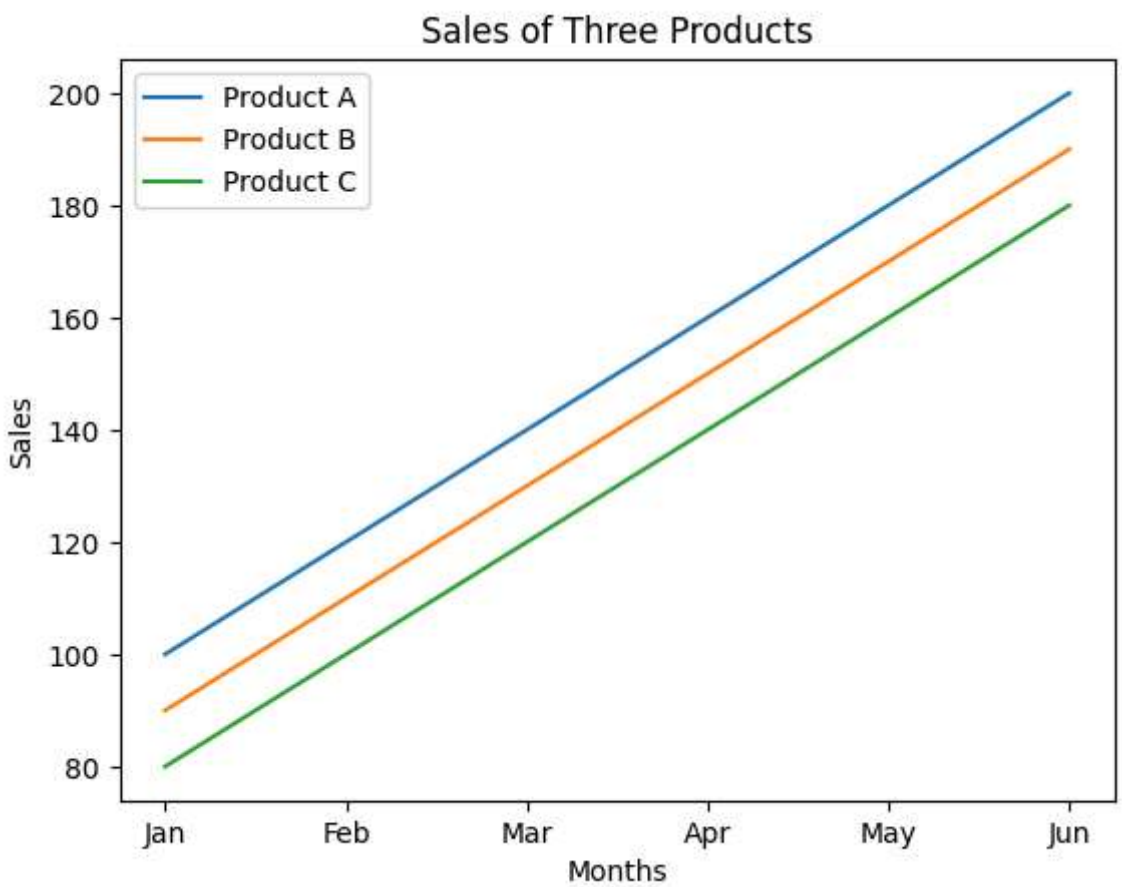
In [1]:
```
# 1. Create a Line chart to display the monthly sales of a company for 12 months.

# import matplotlib library
import matplotlib.pyplot as plt

# Data for months and sales
months = ["Jan","Feb","Mar","Apr","May","Jun","Jul","Aug","Sep","Oct","Nov","Dec"]
sales = [120,150,170,140,180,200,210,190,220,230,250,270]

# Plot line chart
plt.plot(months, sales, marker='o')

# Add title and labels
plt.title("Monthly Sales of Company")
plt.xlabel("Months")
plt.ylabel("Sales")

# Show grid
plt.grid(True)

# Display the chart
plt.show()
```

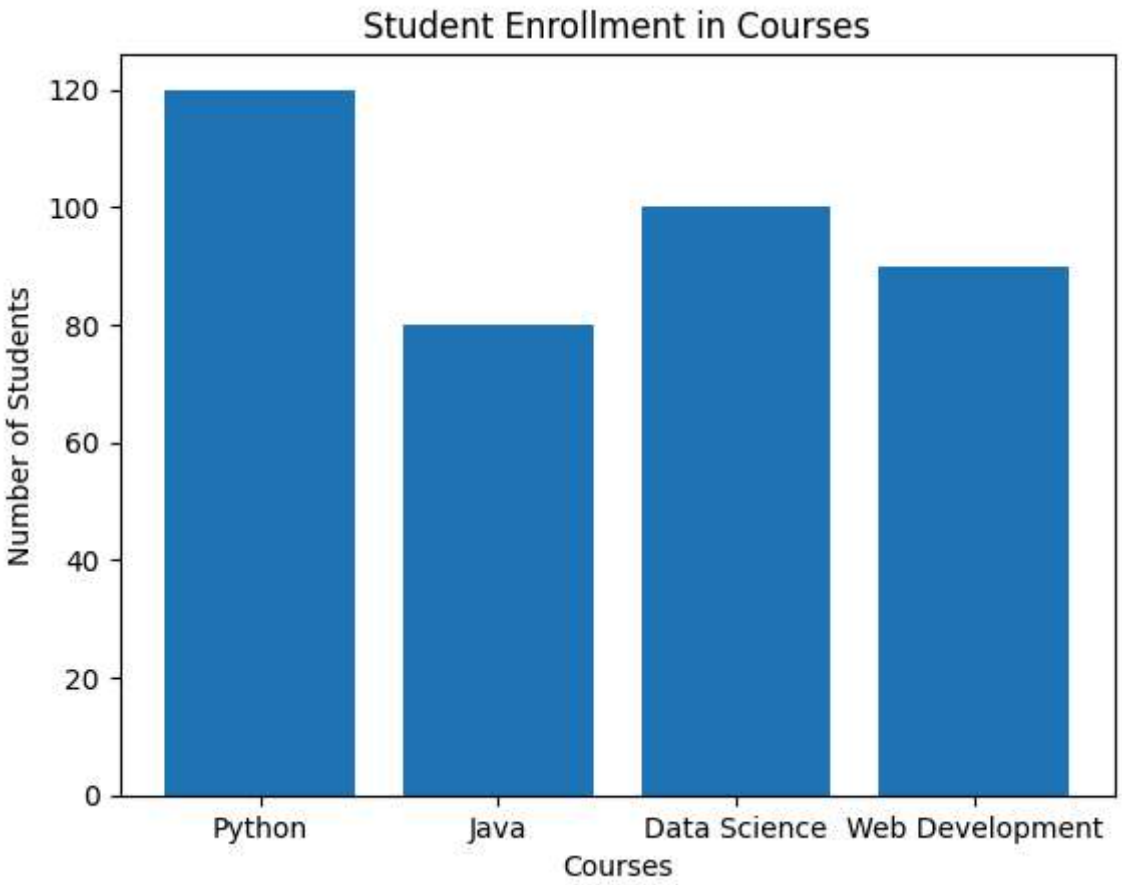
In [2]:
```
# 2. Plot a Multiple Line Chart showing the sales of three products (Product A, B,

import matplotlib.pyplot as plt
```

```
# Months data
months = ["Jan","Feb","Mar","Apr","May","Jun"]

# Sales data for three products
product_A = [100,120,140,160,180,200]
product_B = [90,110,130,150,170,190]
product_C = [80,100,120,140,160,180]

# Plot each product line
plt.plot(months, product_A, label="Product A")
plt.plot(months, product_B, label="Product B")
plt.plot(months, product_C, label="Product C")

# Add title and axis labels
plt.title("Sales of Three Products")
plt.xlabel("Months")
plt.ylabel("Sales")

# Show legend
plt.legend()

# Display chart
plt.show()
```



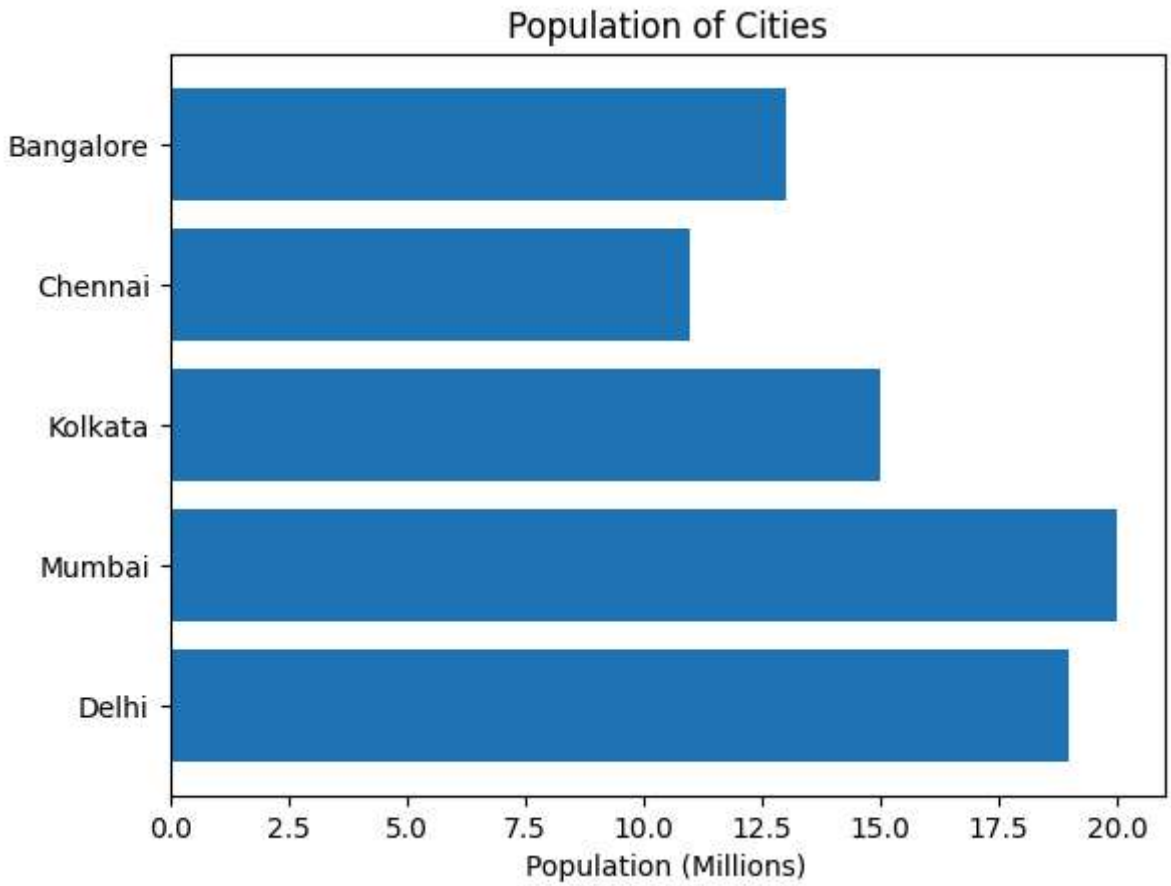
In [3]:
```
# 3. Create a Bar Chart to show the number of students enrolled in different course

import matplotlib.pyplot as plt
```

```
# Courses and number of students
courses = ["Python","Java","Data Science","Web Development"]
students = [120,80,100,90]

# Create bar chart
plt.bar(courses, students)

# Add title and labels
plt.title("Student Enrollment in Courses")
plt.xlabel("Courses")
plt.ylabel("Number of Students")

# Show chart
plt.show()
```



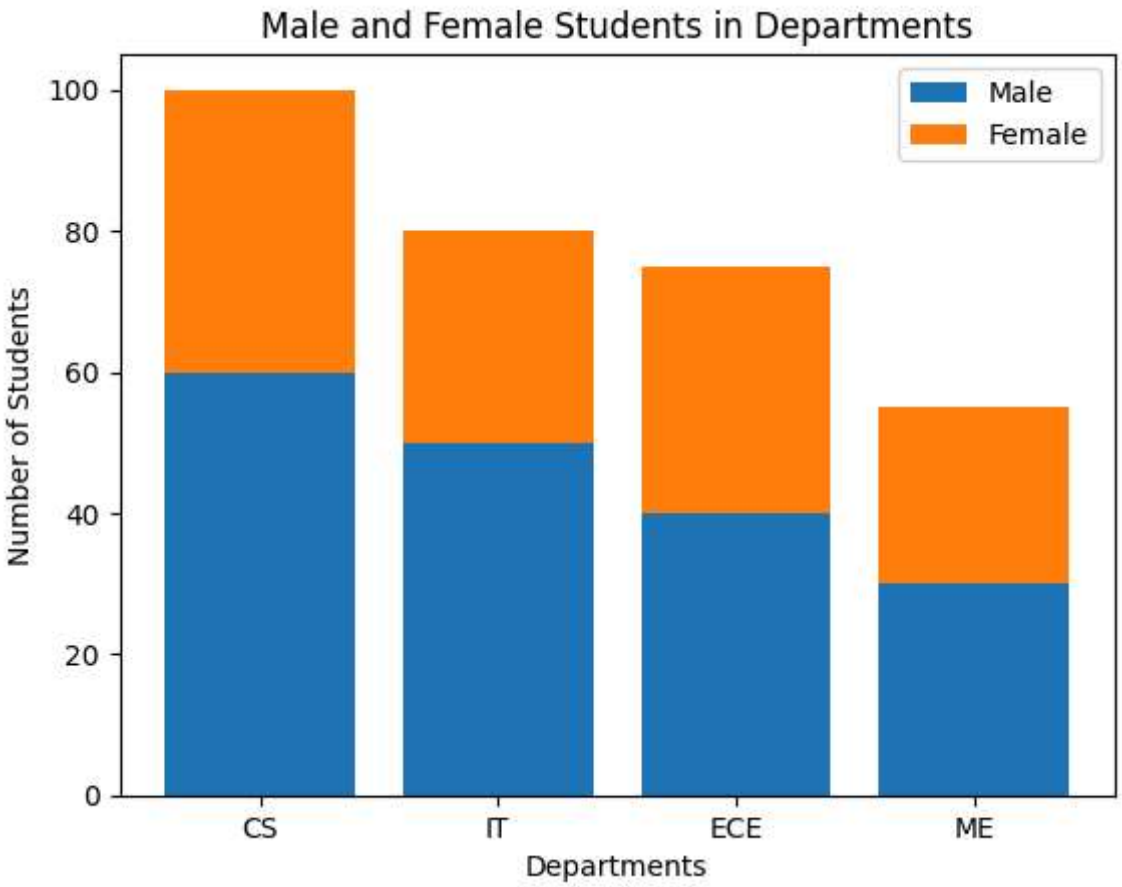
In [6]:
```
# 4. Generate a Horizontal Bar Chart to visualize the population of five different

import matplotlib.pyplot as plt

# Cities and population
cities = ["Delhi","Mumbai","Kolkata","Chennai","Bangalore"]
population = [19,20,15,11,13]

# Create horizontal bar chart
plt.barh(cities, population)

# Add title and label
plt.title("Population of Cities")
```

```
plt.xlabel("Population (Millions)")

# Display chart
plt.show()
```

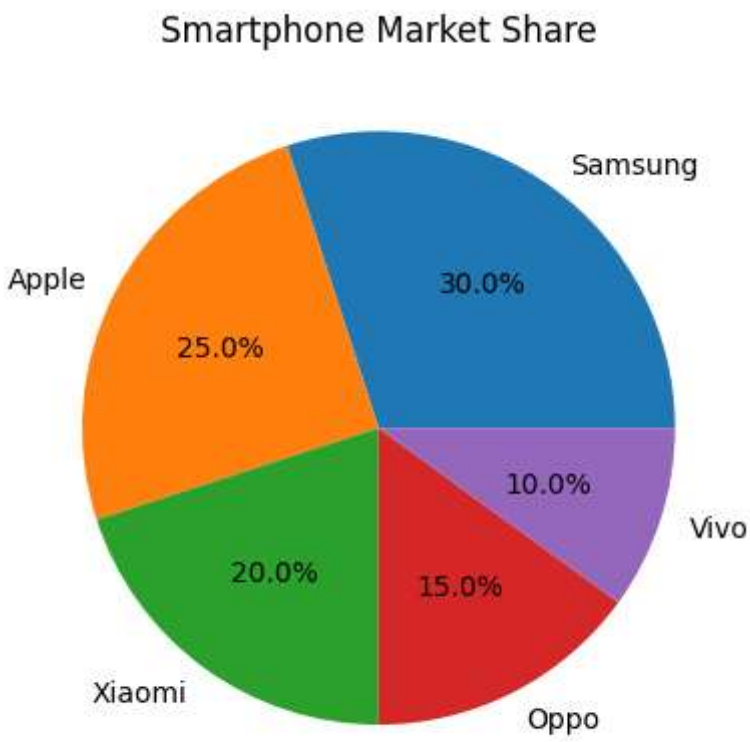


In [7]:
```
# 5. Create a Stacked Bar Chart showing the number of male and female students in f

import matplotlib.pyplot as plt

# Departments
departments = ["CS","IT","ECE","ME"]

# Student data
male = [60,50,40,30]
female = [40,30,35,25]

# Plot male students
plt.bar(departments, male, label="Male")

# Plot female students stacked on male
plt.bar(departments, female, bottom=male, label="Female")

# Add title and labels
plt.title("Male and Female Students in Departments")
plt.xlabel("Departments")
plt.ylabel("Number of Students")

# Show legend
```

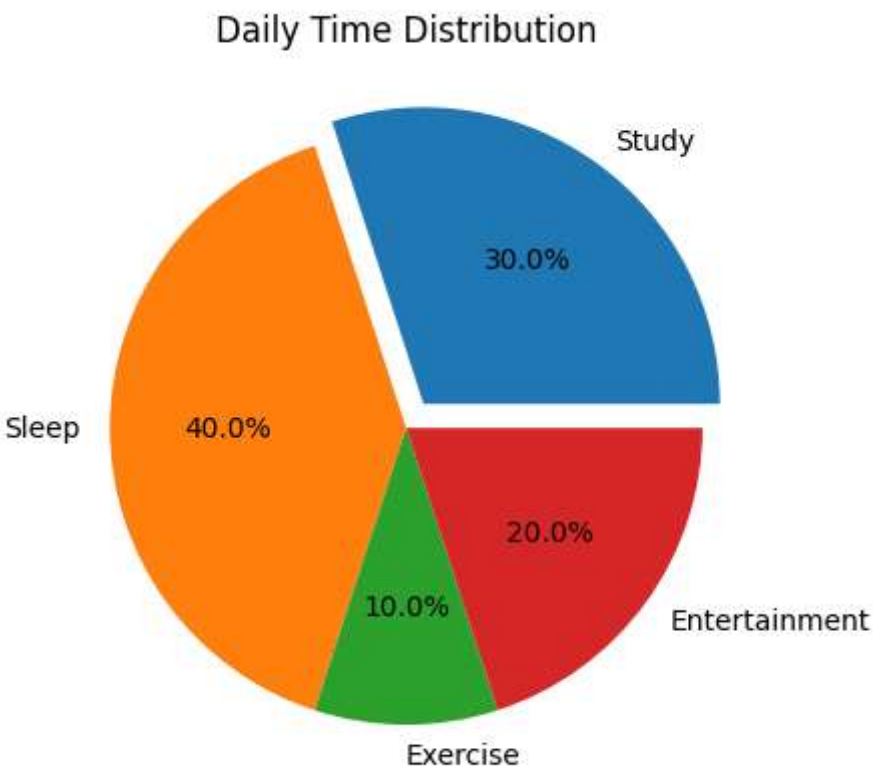
```
plt.legend()

# Display chart
plt.show()
```



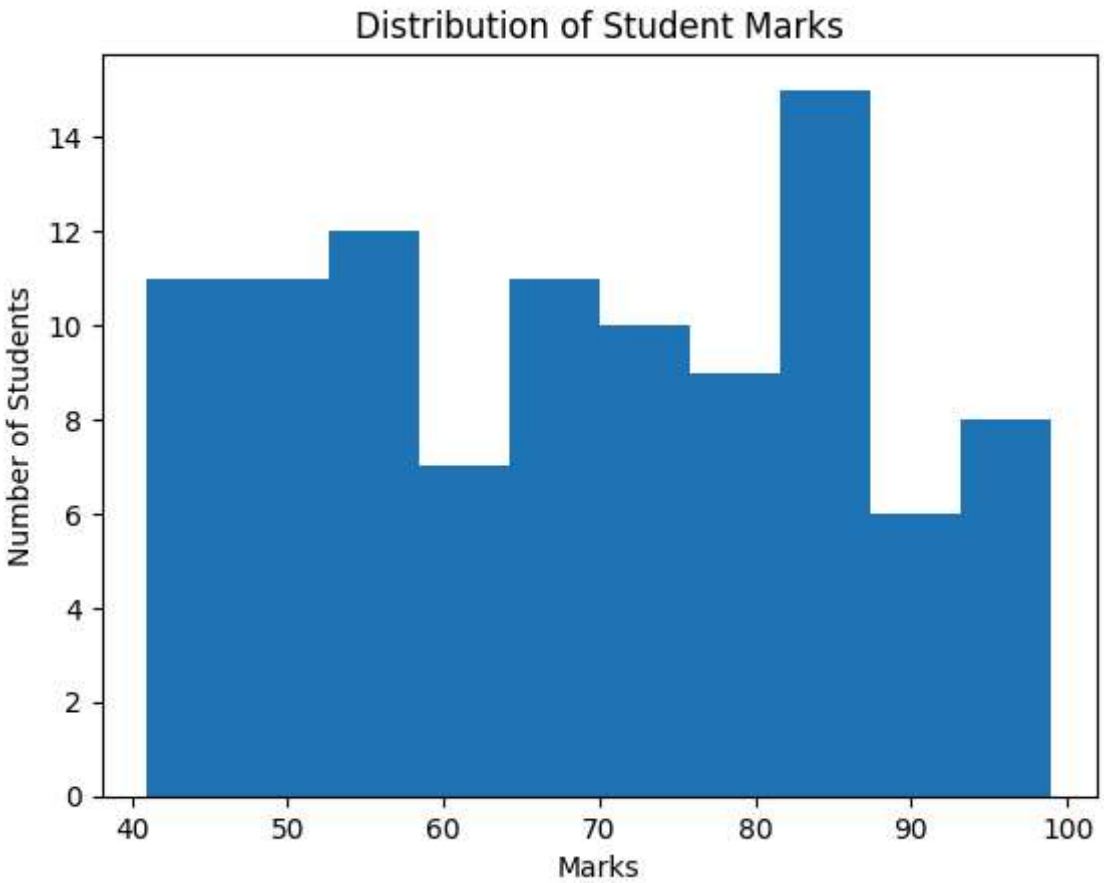
In [8]:
```
# 6. Draw a Pie Chart to represent the market share of five smartphone brands.

import matplotlib.pyplot as plt

# Smartphone brands and market share
brands = ["Samsung","Apple","Xiaomi","Oppo","Vivo"]
share = [30,25,20,15,10]

# Create pie chart
plt.pie(share, labels=brands, autopct='%1.1f%%')

# Add title
plt.title("Smartphone Market Share")

# Show chart
plt.show()
```



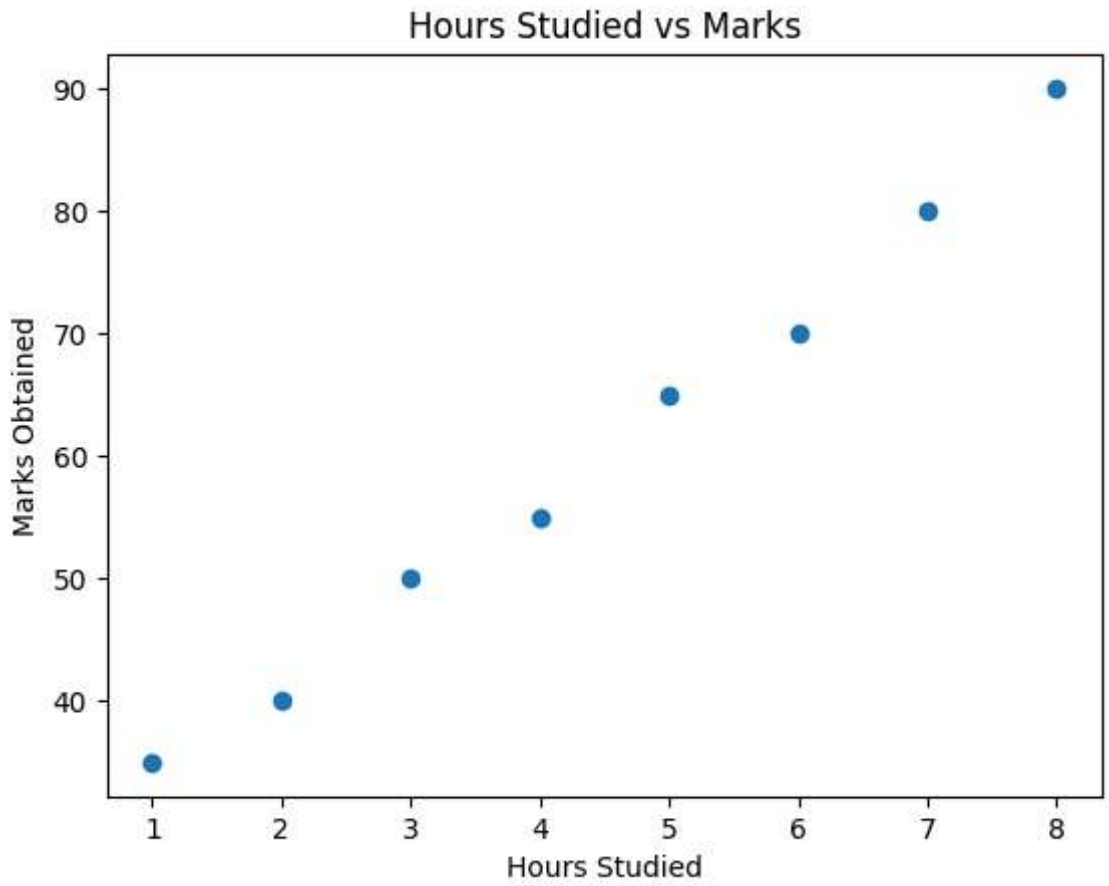
In [9]:
```
# 7. Create an Exploded Pie Chart showing daily time spent on activities like study

import matplotlib.pyplot as plt

# Activities and time spent
activities = ["Study","Sleep","Exercise","Entertainment"]
time = [6,8,2,4]

# Explode first slice (Study)
explode = (0.1,0,0,0)

# Plot exploded pie chart
plt.pie(time, labels=activities, explode=explode, autopct='%1.1f%%')

# Add title
plt.title("Daily Time Distribution")

# Display chart
plt.show()
```



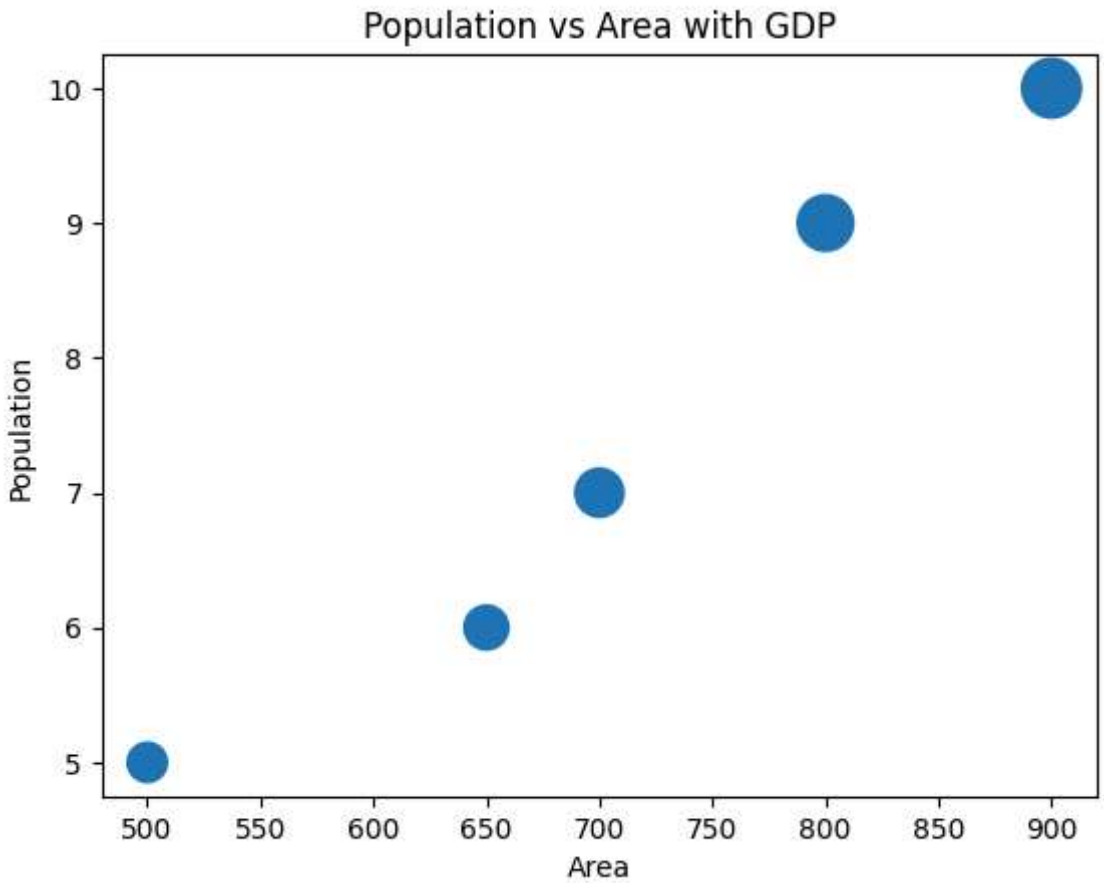
In [10]:
```
# 8. Plot a Histogram to visualize the distribution of marks obtained by 100 studen

import matplotlib.pyplot as plt
import numpy as np

# Generate random marks for 100 students
marks = np.random.randint(40,100,100)

# Create histogram
plt.hist(marks, bins=10)

# Add title and labels
plt.title("Distribution of Student Marks")
plt.xlabel("Marks")
plt.ylabel("Number of Students")

# Show chart
plt.show()
```



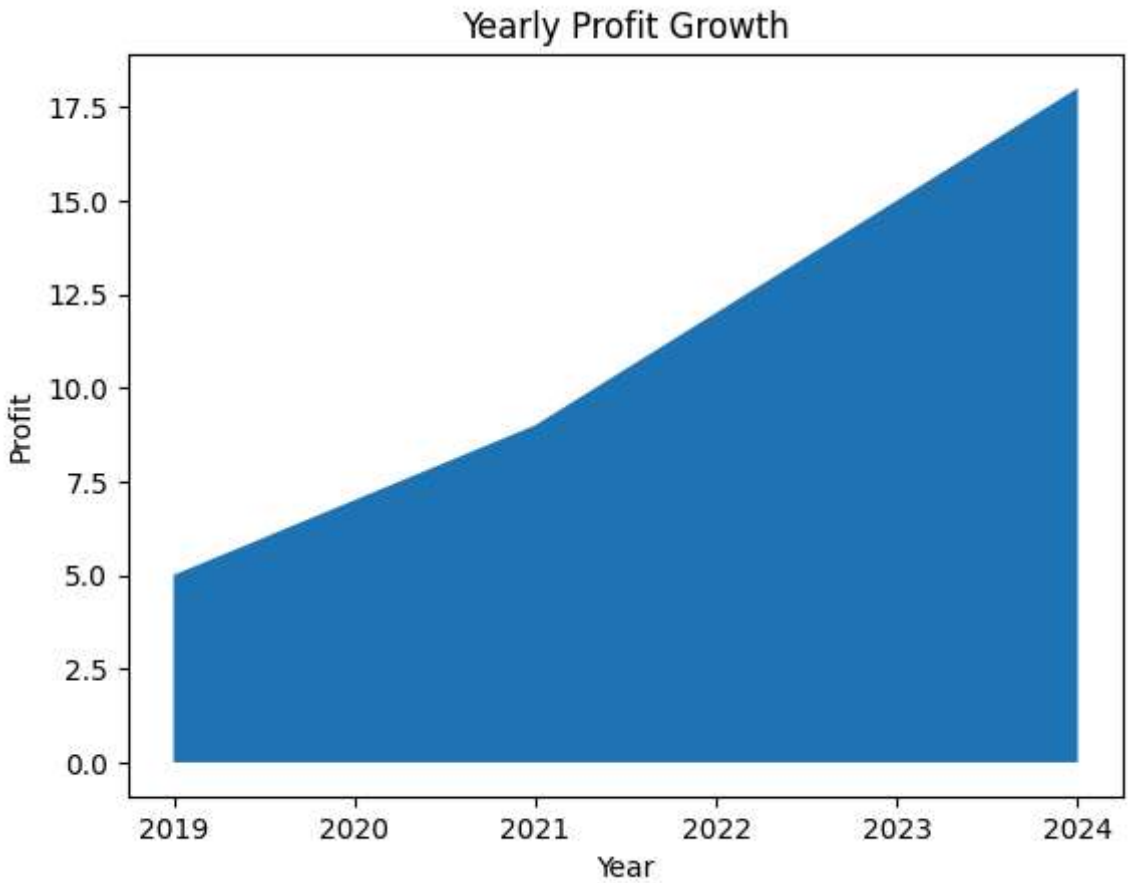
In [11]:
```
# 9. Create a Scatter Plot showing the relationship between hours studied and marks

import matplotlib.pyplot as plt

# Data for hours studied and marks
hours = [1,2,3,4,5,6,7,8]
marks = [35,40,50,55,65,70,80,90]

# Create scatter plot
plt.scatter(hours, marks)

# Add labels and title
plt.title("Hours Studied vs Marks")
plt.xlabel("Hours Studied")
plt.ylabel("Marks Obtained")

# Show chart
plt.show()
```



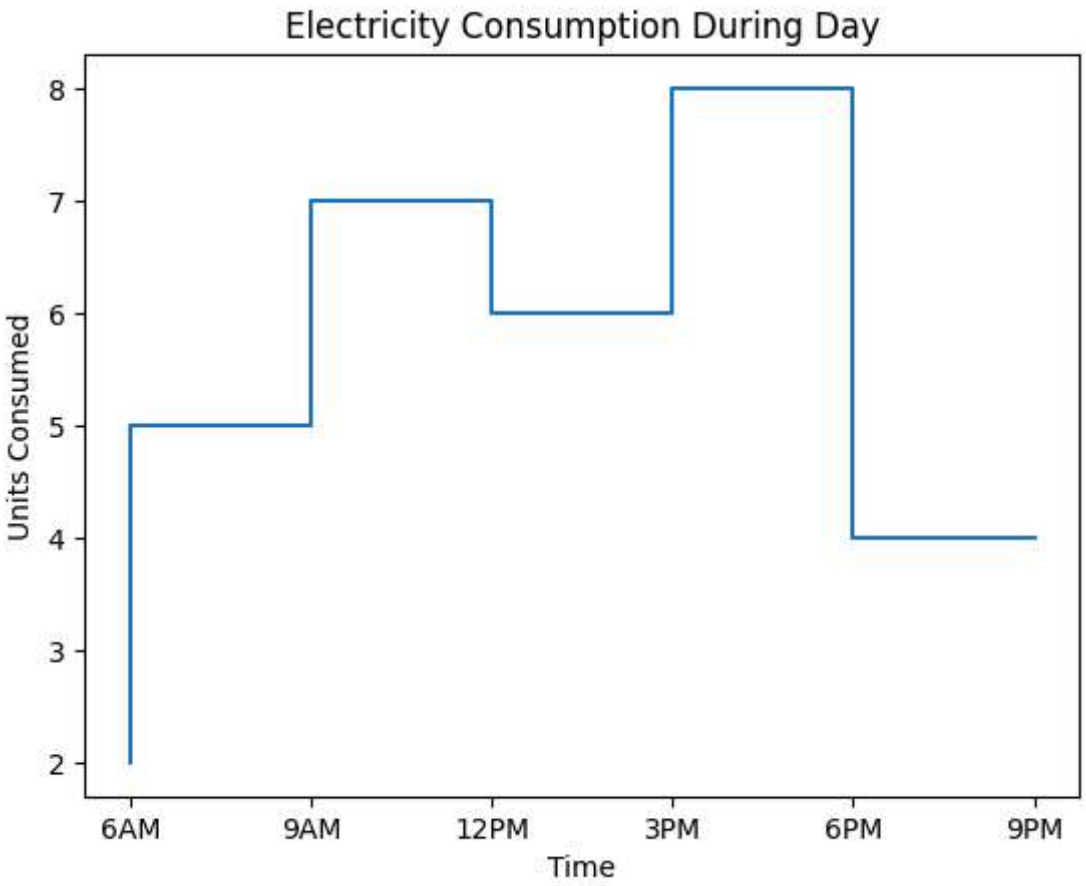
In [12]:
```
# 10. Develop a Bubble Chart showing population vs area of cities, where bubble siz

import matplotlib.pyplot as plt

# City data
cities = ["A","B","C","D","E"]
area = [500,700,800,650,900]
population = [5,7,9,6,10]
gdp = [200,300,400,250,450]

# Bubble chart (size represents GDP)
plt.scatter(area, population, s=gdp)

# Add labels and title
plt.title("Population vs Area with GDP")
plt.xlabel("Area")
plt.ylabel("Population")

# Display chart
plt.show()
```



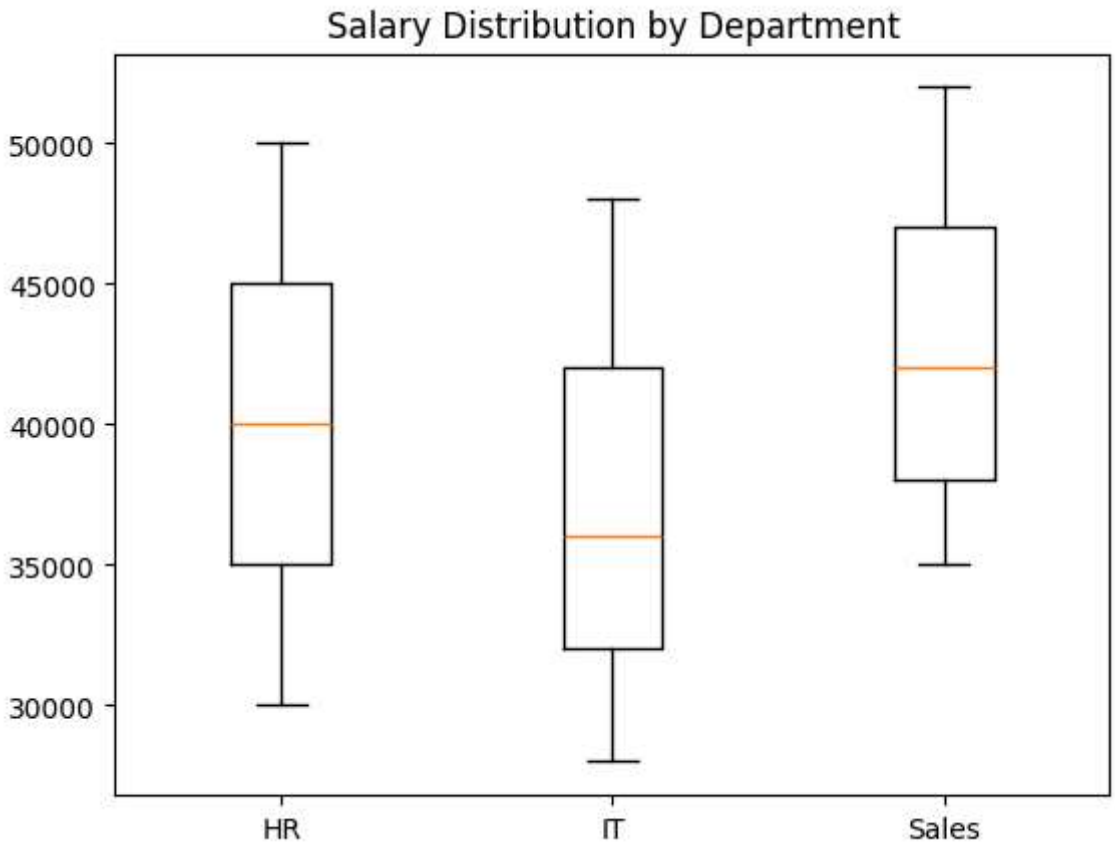
In [14]:
```
# 11. Plot an Area Chart showing the yearly profit growth of a company over 6 years

import matplotlib.pyplot as plt

# Years and profit data
years = [2019,2020,2021,2022,2023,2024]
profit = [5,7,9,12,15,18]

# Create area chart
plt.fill_between(years, profit)

# Add labels
plt.title("Yearly Profit Growth")
plt.xlabel("Year")
plt.ylabel("Profit")

# Display chart
plt.show()
```



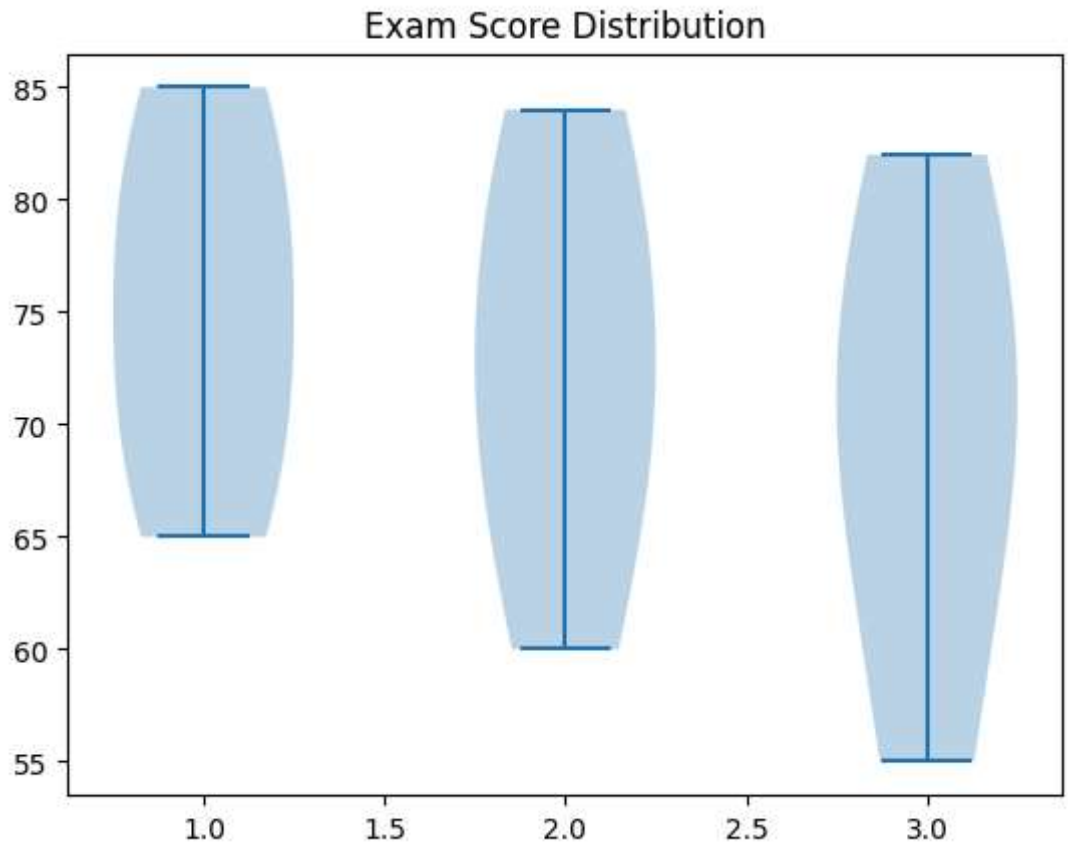
In [15]:
```
# 12. Create a Step Chart to display electricity consumption during different time

import matplotlib.pyplot as plt

# Time intervals and electricity consumption
time = ["6AM","9AM","12PM","3PM","6PM","9PM"]
consumption = [2,5,7,6,8,4]

# Plot step chart
plt.step(time, consumption)

# Add labels and title
plt.title("Electricity Consumption During Day")
plt.xlabel("Time")
plt.ylabel("Units Consumed")

# Show chart
plt.show()
```



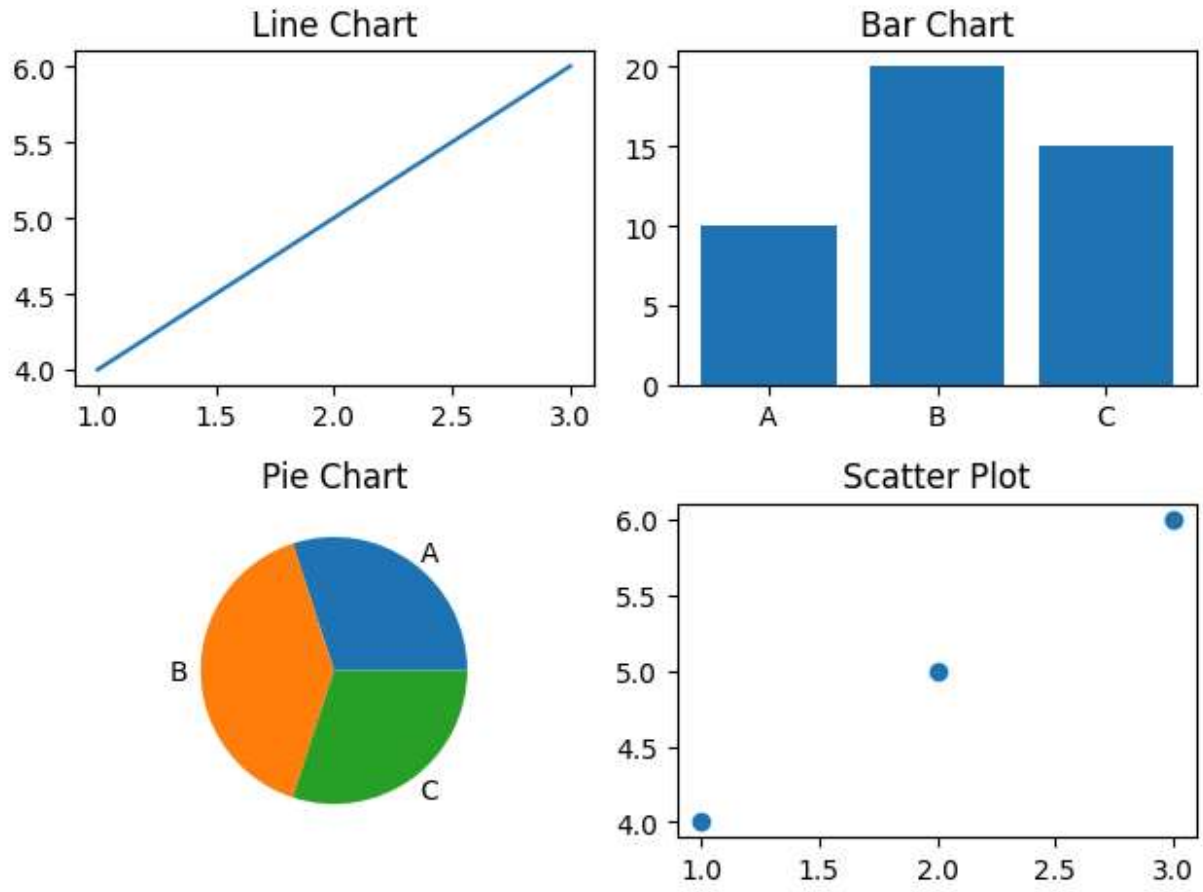
In [17]:
```
# 13. Generate a Box Plot to show the salary distribution of employees in three dif

import matplotlib.pyplot as plt

# Salary data for departments
dept1 = [30000,35000,40000,45000,50000]
dept2 = [28000,32000,36000,42000,48000]
dept3 = [35000,38000,42000,47000,52000]

# Create box plot
plt.boxplot([dept1,dept2,dept3], tick_labels=["HR","IT","Sales"])

# Add title
plt.title("Salary Distribution by Department")

# Display chart
plt.show()
```



In [18]:
```
# 14. Create a Violin Plot to visualize the distribution of exam scores in differen

import matplotlib.pyplot as plt

# Score data for classes
scores = [
    [65,70,75,80,85],
    [60,68,72,78,84],
    [55,65,70,75,82]
]

# Create violin plot
plt.violinplot(scores)

# Add title
plt.title("Exam Score Distribution")

# Display chart
plt.show()
```



In [19]:
```
# 15. Create Subplots in a single figure that display four charts: Line Chart, Bar

import matplotlib.pyplot as plt

# Create figure with 2x2 subplot grid
fig, ax = plt.subplots(2,2)

# Line chart
ax[0,0].plot([1,2,3],[4,5,6])
ax[0,0].set_title("Line Chart")

# Bar chart
ax[0,1].bar(["A","B","C"],[10,20,15])
ax[0,1].set_title("Bar Chart")

# Pie chart
ax[1,0].pie([30,40,30], labels=["A","B","C"])
ax[1,0].set_title("Pie Chart")

# Scatter chart
ax[1,1].scatter([1,2,3],[4,5,6])
ax[1,1].set_title("Scatter Plot")

# Adjust layout
plt.tight_layout()

# Display figure
plt.show()
```


Line Chart
6.0
5.5
5.0
4.5
4.0
1.0
1.5
2.0
2.5
3.0
Bar Chart
20
15
10
5
0
A
B
C
Pie Chart
A
B
C
Scatter Plot
6.0
5.5
5.0
4.5
4.0
1.0
1.5
2.0
2.5
3.0